

Qatar-1b

R-0600

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_231018 · created 2026-07-25T03:28:00+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460235.7366 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.012
Transit depth (Rp/Rs)^2	3.68 +/- 0.47 [%]
Orbital Inclination (inc)	82.31 +/- 0.48
Airmass coefficient 1 (a1)	1.207 +/- 0.014
Airmass coefficient 2 (a2)	-0.1242 +/- 0.0072
Scatter in the residuals of the lightcurve fit is	1.16 %
Best Comparison Star	None
Optimal Aperture	3.55
Optimal Annulus	7.01
Transit Duration (day)	0.0618 +/- 0.0041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.192 $\pm$ 0.012 vs 0.1463 (deviation 3.8 σ)
Duration fit vs published	0.0618 d vs 0.0692 d (deviation 1.79 σ)
Mid-time O–C	6.6 min
Depth SNR	16.0
Residual scatter	1.16 %

Light curve

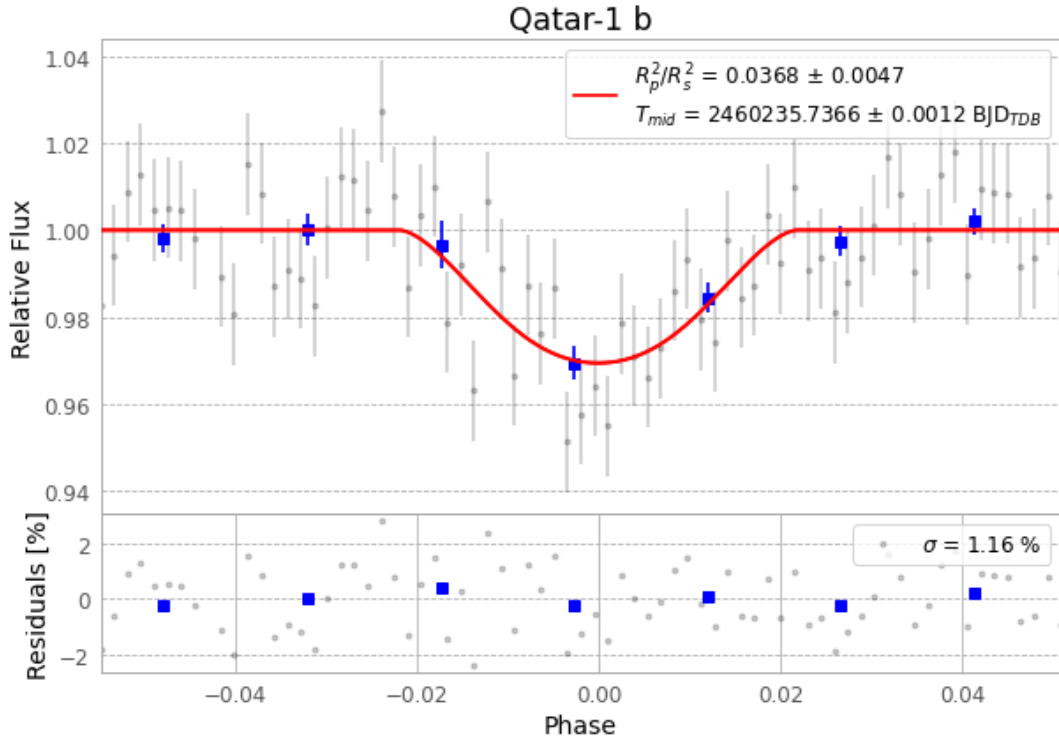


Figure 1 — FinalLightCurve_Qatar-1 b_2023-10-17.png (SHA-256 b9af751c8fb128c5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [282, 189], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 900894841b482a82...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

73 FITS frames (48 MB), Qatar-1231018034516.FITS → Qatar-1231018072115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-231018.log (0003e781ac71...), FinalLightCurve_Qatar-1 b_2023-10-17.png (b9af751c8fb1...), FinalLightCurve_Qatar-1 b_2023-10-17.csv (5223c9a25f9d...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 03:28Z.